Flow Unofficial Docs

Table of contents

1. Flow Unofficial Documentation 3
1.1 Features 3
1.2 Limitations 3
1.3 Security 4
1.4 Other Documentation 4
2. Getting started 5
2.1 Basic example 5
2.2 Updating forms 5
3. Elements/widgets 7
3.1 Layouting elements 7
3.2 Elements 11
3.3 Hypertext markup 34
4. Other features 38
4.1 Unsupported Features 38
4.2 Styling forms 41
4.3 Padding, spacing, and backgrounds 50
4.4 Tooltips 51
4.5 Hiding elements 52
4.6 Using a form as an inventory 53
4.7 Manual positioning 54
4.8 Experimental features 55
4.9 Popovers (highly experimental) 60

1. Flow Unofficial Documentation

Flow is an experimental layout manager and formspec API replacement for Luanti (formerly Minetest): describe the menu, and flow handles the rest. Vaguely inspired by Flutter and GTK.

Luanti's built-in way of describing menus (inventories, chests, dialogs, ...) is called a *formspec*. A formspec is a long string made of elements such as buttons, text fields, and inventory slots, and you have to give every element its exact position and size by hand. Flow lets you describe a menu with Lua tables instead, positions and sizes everything for you, and generates the formspec itself, so you never have to write one.

Online tutorial/demo (source)

1.1 Features

1.1.1 Layouting

- Place elements without coordinates: flow works out every position for you.
- Automatic layouting using `HBox` and `VBox` containers
- Some elements have an automatic size.
- Elements can grow to fill spare space when you set `expand = true`.
- Elements which get their size based on their label length automatically become larger/smaller to fit long translations[1].

1.1.2 Other features

- Form names are handled for you. They are still used internally, however they are hidden from the API.
- Form state is tracked for you.
- Values of fields, scrollbars, checkboxes, etc are remembered when redrawing a form and are automatically applied.
- Includes an inspector mod to help with developing and debugging forms.
- Some common security issues with formspec input handling are mitigated.

1.2 Limitations

- This mod doesn't support all of the features that regular formspecs do. See Unsupported Features for a list.

1.3 Security

Flow ignores potentially malicious formspec input from clients, such as buttons or fields that haven't been shown to the client, out-of-bounds dropdown selections, and newlines in `Field` elements (where it's impossible to enter a newline without pasting it in).

1.4 Other Documentation

The official documentation is available at https://luk3yx.gitlab.io/minetest-flow/. Some code snippets have a "run" button which will open them in a web-based playground, not all of these will work properly as the playground doesn't support all formspec elements.

---

1. This isn't perfect, the actual size of labels depends on the client's font, DPI, window size, and formspec-related scaling settings. ↩

2. Getting started

A *form* is a menu that is shown to a player. It is made of *elements* (also called widgets), such as labels, buttons, text fields, and inventory slots, which are arranged by layout elements like `gui.VBox`. You describe a form with a *build function* that returns the elements, and flow works out the positions and sizes and shows the form for you. See Elements for what each element does.

2.1 Basic example

See `example.lua` for a more comprehensive example which demonstrates how layouting and alignment works.

```
local gui = flow.widgets -- local recommended for direct referencing
local STARTING_SELECTION = 3 -- change and rerun to see the difference

local function build_vbox(player_obj, ctx)
    return gui.VBox {
        gui.Label { label = "Here is a dropdown:" },
        gui.Dropdown {
            name = "cool_dropdown",
            items = { 'First item', 'Second item', 'Third item' },
            index_event = true,
            selected_idx = ctx.value or 1
        },
        gui.Button {
            label = "Get dropdown index",
            on_event = function(player, ctx)
                local idx = ctx.form.cool_dropdown or ctx.value or 1
                core.chat_send_player(player:get_player_name(),
                    ("You have selected item #%d!"):format(idx))
            end,
        }
    }
end

local form = flow.make_gui(build_vbox)
local player_obj = core.get_player_by_name("playground")

form:show(player_obj, { value = STARTING_SELECTION }) -- second argument is optional
```

2.2 Updating forms

If some data displayed inside a form changes (for example a timer or progress indicator), you can use `form:update` to update the form without resetting `ctx` or showing the form again if the player has closed it.

Luanti has no way to change a form that is already open, so flow has to show the whole form again. Because of this, players may lose text typed into fields that hasn't been sent to the server (which happens when they press a button or Enter) when `form:update` is called.

```
-- Re-shows the form for one player if they have the form open
my_gui:update(player)

-- Re-shows the form for all players that have the form open and where
-- ctx.test == 123
my_gui:update_where(function(player, ctx)
    return ctx.test == 123
end)

-- Re-shows the form for all players with the "server" privilege
my_gui:update_where(function(player, ctx)
    return core.check_player_privs(player, "server")
end)

-- Re-shows the form for all players with the form open
my_gui:update_where(function() return true end)
```

Inside an `on_event` handler, you can use `return true` instead.

```
gui.Button {
    label = "Update form",
    on_event = function(player, ctx)
        return true
    end,
}
```

If you're using a form as a HUD, you must call `form:show_hud` to update it.

3. Elements/widgets

3.1 Layouting elements

You should do `local gui = flow.widgets` in your code to improve readability. All examples will assume that this line exists.

These elements control how other elements are arranged. Most of them draw nothing themselves: they place the elements inside them, work out how big everything is, and share out any spare space. Elements that draw something or accept input (labels, buttons, fields, ...) are described in Elements. To place elements at exact coordinates instead, see Manual positioning.

Warning: The below elements (except `gui.Spacer`) do not support `w` and `h` properties and will break in weird ways if you try and use them. Use `min_w` and `min_h` instead.

3.1.1 `gui.VBox`

A vertical box, similar to a VBox in GTK. Elements inside a VBox are stacked vertically.

```
gui.VBox {
    -- These elements are documented later on.
    gui.Label { label = "I am a label!" },

    -- The second label will be positioned underneath the first one.
    gui.Label { label = "I am a second label!" },
}
```

Elements inside boxes have a spacing of 0.2 between them. To change this, you can add `spacing = <number>` to the box definition. For example, `spacing = 0` will remove all spacing between the elements.

3.1.2 `gui.HBox`

Like `gui.VBox` but stacks elements horizontally instead.

```
gui.HBox {
    -- These elements are documented later on.
    gui.Label { label = "I am a label!" },

    -- The second label will be positioned to the right of first one.
    gui.Label { label = "I am a second label!" },

    -- You can nest HBox and VBox elements
    gui.VBox {
        gui.Image { w = 1, h = 1, texture_name = "default_dirt.png", align_h = "center" },
        gui.Label { label = "Dirt", expand = true, align_h = "center" },
    }
}
```

3.1.3 `gui.ScrollableVBox`

Similar to `gui.VBox`, but shows its contents in an area of a fixed size that the player can scroll, and automatically adds a scrollbar next to it. Anything outside the area is hidden until it is scrolled into view. The area is `w` by `h`, both are optional: the width defaults to the width of the contents and the height defaults to the height of the contents, up to a maximum of 5.

On clients too old to support scrolling areas (before Luanti 5.4), flow shows one page of the contents at a time instead, with `<` and `>` buttons to change page.

To change how the scrollbar looks, set `custom_scrollbar` to a table of parameters for `gui.Scrollbar`, for example `custom_scrollbar = {w = 0.8}` to make it wider.

```
gui.ScrollableVBox {
    -- A name must be provided for ScrollableVBox elements. You don't
    -- have to use this name anywhere else, it just makes sure flow
    -- doesn't mix up scrollbar states if one gets removed or if the
    -- order changes.
    name = "vbox1",

    -- Specifying a height is optional but is probably a good idea.
    -- If you don't specify a height, it will default to
    -- min(height_of_content, 5).
    h = 10,

    -- These elements are documented later on.
    gui.Label { label = "I am a label!" },

    -- The second label will be positioned underneath the first one.
    gui.Label { label = "I am a second label!" },
}
```

3.1.4 `gui.Spacer`

A "flexible space" element that expands by default. Example usage:

```
gui.HBox {
    -- These buttons will be on the left-hand side of the screen
    gui.Button { label = "Cancel" },
    gui.Button { label = "< Back" },

    gui.Spacer {},

    -- These buttons will be on the right-hand side of the screen
    gui.Button { label = "Next >" },
    gui.Button { label = "Confirm" },
}
```

I advise against using spacers when `expand = true` and `align = ...` would work just as well since spacers are implemented hackily and won't account for some special cases.

You can replicate the above example without spacers, however the code doesn't look as clean:

```
gui.HBox {
    -- These buttons will be on the left-hand side of the screen
    gui.Button { label = "Cancel" },
    gui.Button { label = "< Back", expand = true, align_h = "left" },

    -- These buttons will be on the right-hand side of the screen
    gui.Button { label = "Next >" },
    gui.Button { label = "Confirm" },
}
```

You should not use spacers to center elements as it creates unnecessary boxes, and labels may be slightly off-center (because label widths depend on screen size, DPI, etc and this code doesn't trigger the centering hack):

```
-- This is bad!
gui.HBox {
    gui.Spacer {},
    gui.Label { label = "I am not properly centered!" },
    gui.Spacer {},
}
```

You should do this instead:

```
gui.Label { label = "I am centered!", align_h = "center" },
```

This applies to other elements as well, because using HBox and Spacer to center elements creates unnecessary containers.

3.1.5 `gui.Nil`

A tool to allow for ternary-ish conditional widgets:

```
local form = flow.make_gui(function(player, ctx)
    local the_boolean = false
    return gui.VBox {
        gui.Label { label = "The box below is only present if the boolean is truthy" },
        the_boolean and gui.Box { w = 1, h = 1, color = "#FF0000" } or gui.Nil {},
    }
end)
```

Use sparingly, flow still has to process each `Nil` object to be able to know to remove it, and thus could still slow things down. The fastest element is one that doesn't exist, and thus doesn't need processing.

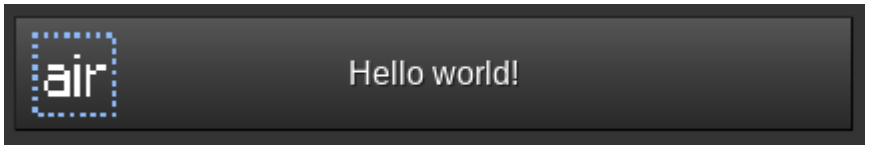
3.1.6 `gui.Stack`

This container element places its children on top of each other. All child elements are expanded in both directions.

Note that some elements (such as centred labels) won't pass clicks through to the element below them.

Example

```
gui.Stack {
    min_w = 10,
    gui.Button { label = "Hello world!" },
    gui.Image { w = 1, h = 1, texture_name = "air.png", padding = 0.2, align_h = "left" },
}
```



3.1.7 `gui.PaginatedVBox`

Like `gui.ScrollableVBox`, but splits the contents into pages instead of scrolling. Only one page is shown at a time, with a `<` button, a "Page 1 of 3" label, and a `>` button underneath to change page. It needs a unique `name`, and `w` and `h` work the same way as for `gui.ScrollableVBox`. Elements are never cut in half: an element that does not fit on a page starts the next one.

`gui.ScrollableVBox` uses this automatically on clients that cannot scroll.

3.1.8 `gui.Padding` (deprecated)

Wraps a single element to add padding around it. It is deprecated and logs a warning, use the `padding` parameter of the element itself instead, see Padding.

3.2 Elements

This page describes the elements (widgets) that draw something or accept input. Elements that only arrange other elements (`gui.VBox` , `gui.HBox` , `gui.Stack` , ...) are described in Layouting elements.

Every element is a function in `flow.widgets` that takes a table of parameters:

```
local gui = flow.widgets

gui.Label { label = "Hello world!" }
```

The parameter lists below are generated automatically. Parameters marked `-- Optional` can be left out.

3.2.1 Concepts used on this page

Sizes

Sizes are measured in inventory slots: `1` is the width and height of one item slot. There is no hidden padding or spacing, so two boxes that are `1` wide and placed next to each other with `spacing = 0` touch. You never give an element a position, boxes place their children and flow works out the total size of the form.

Elements that show text, such as labels, buttons, and fields, are sized to fit their text unless you give them a `w` and `h` . Elements that show something without a natural size, such as images, boxes, models, and tables, need an explicit `w` and `h` (their parameter lists below do not say `-- Optional`). Flow can only estimate the width of text because the real width depends on the player's font, screen size, and scaling settings, so avoid layouts that rely on exact text widths.

Names and values

Many elements take a `name` . Input elements (fields, checkboxes, dropdowns, lists of text, tables, ...) with a `name` have their current value stored in `ctx.form[name]` and flow remembers it when the form is redrawn:

Element	Value in `ctx.form[name]`
`gui.Field` , `gui.Pwdfield` , `gui.Textarea`	Text (string)
`gui.Checkbox`	`true` or `false`
`gui.Dropdown`	Text of the selected item, or its number if `index_event = true`
`gui.Textlist` , `gui.Table`	Number of the selected row (first row is `1`)

Until the player changes something, the value is the element's initial value (`default` , `selected` , `selected_idx`), or `""` , `false` , or `1` if you did not give one. Text fields only send their text to the server when the player presses a button or Enter, so `ctx.form` can be out of date until then.

If your build function reads `ctx.form.some_name` , flow redraws the form automatically whenever the player changes that value (unless the element has its own `on_event`). Buttons do not store a value.

Events

Elements that the player can use accept an `on_event` function, which is called with the player and `ctx` . Return `true` from it to redraw the form. A `name` is optional if you use `on_event` , flow makes one up.

Colors

Colors are strings. `"#RGB"` and `"#RRGGBB"` are fully opaque, while `"#RGBA"` and `"#RRGGBBAA"` end with an alpha value, where `00` is fully transparent and `FF` is fully opaque. Color names also work (all the names from CSS, such as `"red"` and `"darkgreen"`), and you can add an alpha value after a `#` , for example `"darkgreen#08"` .

Most text can also be coloured by putting color escape sequences in it, such as the ones made by `core.colorize()` . Dropdown items and tab captions are an exception, they cannot be coloured.

Textures and items

A texture name is the file name of an image from the `textures` folder of any loaded mod, such as `"default_dirt.png"` . An item name is the registered name of an item or node, such as `"default:dirt"` .

Parameters shared by all elements

Every element, including layouting elements, accepts these parameters in addition to its own:

Parameter	Description
`padding`	Space around the element, see Padding.
`visible`	Set to `false` to hide the element but keep its space, see Hiding elements.
`expand`	Set to `true` to make the element grow to fill spare space in the `gui.HBox` or `gui.VBox` it is in. If several elements expand, they share the spare space equally.
`align_h`, `align_v`	Where the element goes when there is more room than it needs: `"start"`, `"end"`, `"center"`, `"fill"` (stretch), or `"auto"` (the default, flow picks one). `"left"` and `"top"` mean `"start"`, `"right"` and `"bottom"` mean `"end"`, and `"center"` means `"center"`.
`min_w`, `min_h`	The smallest size of the element, for elements that size themselves.
`tooltip`	Text to show when the player hovers over the element, see Tooltips.
`style`	Inline styles, see Styling.
`on_event`	A function to call when the player uses the element, see Events above.

Nine-slice images

Images and image-like backgrounds can optionally be drawn as a "nine-slice". The picture is cut into a 3x3 grid, where the corners keep their size, the edges are stretched in one direction, and the middle is stretched in both. This lets one small texture make a panel or border of any size without distorting its corners. The `middle_*` parameters describe where the cuts are, in texture pixels (whole numbers):

- `middle_x` only: the cuts are `middle_x` pixels from every side.
- `middle_x` and `middle_y`: `middle_x` pixels from the left and right sides and `middle_y` pixels from the top and bottom.
- All four: the middle rectangle starts at `middle_x, middle_y` and ends at `middle_x2, middle_y2`. Negative `middle_x2` and `middle_y2` values are measured from the far (right/bottom) edge of the texture.

3.2.2 Elements

`gui.AnimatedImage`

Shows an image that plays an animation. The texture must contain every frame stacked on top of each other (the first frame at the top), with all frames the same size. `w` and `h` are the size of one displayed frame.

`middle_*` parameters turn on nine-slice drawing, see Nine-slice images.

Example

```
gui.AnimatedImage {
    w = 1,
    h = 2,

    -- Only useful for styling.
    name = "my_animated_image", -- Optional

    -- Texture that contains all frames.
    texture_name = "texture.png",

    -- Number of frames in the texture.
    frame_count = 3,

    -- Milliseconds each frame is shown for. 0 means no animation.
    frame_duration = 4,

    -- Frame to start on, the first frame is 1. Default 1.
    frame_start = 5, -- Optional
    middle_x = 6, -- Optional
    middle_y = 7, -- Optional
    middle_x2 = 8, -- Optional
    middle_y2 = 9, -- Optional
}
```

`gui.Background`

Draws a picture behind everything else in the form: all backgrounds are drawn before any other element, whatever order they are in. The picture (`texture_name`) is stretched to `w` by `h` . If `auto_clip` is `true` , it is cut off at the edge of the form instead and `w` and `h` are ignored.

You usually do not need this element, set `bgimg` on a container or the root element instead (see Padding). Inside a `gui.Stack` , a `gui.ScrollableVBox` , or an embedded form, flow draws backgrounds without `auto_clip` as ordinary images.

Example

```
gui.Background {
    w = 1,
    h = 2,

    -- Picture to draw.
    texture_name = "texture.png",

    -- Cut off at the form edge, ignore w and h.
    auto_clip = false, -- Optional
}
```

`gui.Background9`

Like `gui.Background`, but the picture is drawn as a nine-slice so that it can be stretched to any size without distorting its corners.

Example

```
gui.Background9 {
    w = 1,
    h = 2,

    -- Picture to draw.
    texture_name = "texture.png",

    -- Cut off at the form edge, ignore w and h.
    auto_clip = false,

    -- Nine-slice cut, see "Nine-slice images".
    middle_x = 3,
    middle_y = 4, -- Optional
    middle_x2 = 5, -- Optional
    middle_y2 = 6, -- Optional
}
```

`gui.Bgcolor`

Sets the colors behind the form. This does the same thing as the `bgcolor`, `fbgcolor`, and `bg_fullscreen` options of the root element, which are explained in Experimental features and are easier to use. `fullscreen` can be `false`, `true`, `"both"`, or `"neither"`.

Example

```
gui.Bgcolor {
    -- Background color of the form.
    bgcolor = "#FF0000",

    -- false, true, "both", or "neither".
    fullscreen = false, -- Optional

    -- Color of the area behind the form.
    fbgcolor = "#FF0000", -- Optional
}
```

`gui.Box`

Draws a solid rectangle of one color. Boxes have no natural size, so give them a `w` and `h`.

Unlike everywhere else, a hexadecimal color without an alpha value, such as `"#FF0000"`, is drawn semi-transparent here. Add the alpha value (`"#FF0000FF"`) to make it opaque.

Example

```
gui.Box {
    w = 1,
    h = 2,

    -- Fill color.
    color = "#FF0000",
}
```

`gui.Button`

A clickable button showing `label`. Your `on_event` function is called when it is clicked. If you do not set `w` and `h`, the button is sized to fit the label (but never smaller than 1.6 by 0.8).

Buttons do not store a value in `ctx.form`.

Example

```
gui.Button {
    w = 6, -- Optional
    h = 2, -- Optional
    name = "my_button", -- Optional
    label = "Hello world!", -- Optional
}
```

`gui.ButtonExit`

A button that closes the form when it is clicked. Your `on_event` function is still called before the form closes. In all other respects it works like `gui.Button`.

Example

```
gui.ButtonExit {
    w = 6, -- Optional
    h = 2, -- Optional
    name = "my_button_exit", -- Optional
    label = "Hello world!", -- Optional
}
```

`gui.ButtonURL`

A button that, when clicked, offers the player the choice of opening `url` in their web browser. `url` must start with `http://` or `https://`. Your `on_event` function is also called. In all other respects it works like `gui.Button`.

Example

```
gui.ButtonURL {
    w = 6, -- Optional
    h = 2, -- Optional
    name = "my_button_url", -- Optional
    label = "Hello world!", -- Optional

    -- Web address to offer to open.
    url = "Hello world!",
}
```

`gui.ButtonUrlExit`

Like `gui.ButtonURL`, but the form also closes when the button is clicked.

Example

```
gui.ButtonUrlExit {
    w = 6, -- Optional
    h = 2, -- Optional
    name = "my_button_url_exit", -- Optional
    label = "Hello world!", -- Optional

    -- Web address to offer to open.
    url = "Hello world!",
}
```

`gui.Checkbox`

A tick box with a text `label`. `selected` sets whether it starts ticked. If you give it a `name`, `ctx.form[name]` is `true` or `false` and is updated as soon as the player ticks or unticks it.

Checkboxes are sized automatically and do not accept `w` or `h`.

Example

```
gui.Checkbox {
    name = "my_checkbox", -- Optional
    label = "Hello world!", -- Optional
    selected = false, -- Optional
}
```

`gui.Container`

A group of elements that you position yourself with `x` and `y` instead of letting flow lay them out. See Manual positioning.

Example

```
gui.Container {
    w = 6, h = 2, -- Optional
    { type = "box", x = 0, y = 0, w = 1, h = 1, color = "red" },
    { type = "label", x = 1, y = 1.5, label = "Hello world!" },
}
```

`gui.Dropdown`

A drop-down list that shows the currently selected item and lets the player pick another one from `items` (a list of strings). `selected_idx` is the number of the item that is selected at first (the first item is `1`).

`ctx.form[name]` is the text of the selected item. If `index_event = true` it is the number of the selected item instead, which is easier to work with when two items have the same text or when the text is translated.

If you do not set `w` and `h`, the dropdown is sized to fit its longest item (but never smaller than 2 by 0.8). Item text cannot be coloured.

Example

```
gui.Dropdown {
    w = 2, -- Optional
    h = 2, -- Optional
    name = "my_dropdown", -- Optional

    -- List of choices.
    items = { "Hello world!", ... },

    -- Number of the item selected at first.
    selected_idx = 3, -- Optional

    -- Store the item number in ctx.form instead of its text.
    index_event = false, -- Optional
}
```

`gui.Field`

A single-line text box. `label`, if set, is shown above the box (flow leaves room for it automatically). `default` is the text it starts with.

`ctx.form[name]` is the text in the box. Newlines and other control characters are removed from it (a player can only get them into a field by pasting). The text is sent to the server when the player presses a button or Enter. To react to Enter, see Handling enter keypresses in fields.

If you do not set `w` and `h`, the field is sized to fit its default text and label (but never smaller than 3 by 0.8).

Example

```
gui.Field {
    w = 6, -- Optional
    h = 2, -- Optional
    name = "my_field", -- Optional
    label = "Hello world!", -- Optional

    -- Text the field starts with.
    default = "Hello world!", -- Optional

    -- Called when the player presses Enter.
    on_key_enter = function(player, ctx) end, -- Optional

    -- Close the form on Enter. Default false.
    close_on_enter = false, -- Optional

    -- Make "Done" on the mobile on-screen keyboard act like Enter.
    enter_after_edit = false, -- Optional
}
```

`gui.FieldCloseOnEnter`

Sets whether pressing Enter in the field with this `name` closes the form. You normally do not need this element, as `gui.Field` has a `close_on_enter` parameter (and by default Enter does not close the form).

Example

```
gui.FieldCloseOnEnter {
    -- Name of the field.
    name = "my_field_close_on_enter",
    close_on_enter = false,
}
```

`gui.FieldEnterAfterEdit`

Experimental, and only affects Android. If `enter_after_edit` is `true`, pressing "Done" in the text input dialog of the field with this `name` counts as pressing Enter. You normally do not need this element, as `gui.Field` has an `enter_after_edit` parameter.

Example

```
gui.FieldEnterAfterEdit {
    -- Name of the field.
    name = "my_field_enter_after_edit",
    enter_after_edit = false,
}
```

`gui.Hypertext`

Shows text that can be formatted (bold, colors, different sizes, images, ...) and can contain clickable links. The text is written in a small HTML-like language, which is described in Hypertext markup. `w` and `h` are the size of the area that the text is wrapped to fit in.

If the markup contains `<action name=...>` links, `ctx.form[name]` is set to `"action:"` followed by the action name when the player clicks one. This behaviour is experimental and may change.

Example

```
gui.Hypertext {
    w = 6,
    h = 2,

    -- Receives the clicked action, see above.
    name = "my_hypertext", -- Optional

    -- Formatted text, see "Hypertext markup".
    text = "Hello world!",
}
```

`gui.Image`

Shows a picture, stretched to fit `w` by `h`. If you want to use one small texture for pictures of many sizes, use the `middle_*` parameters, see Nine-slice images.

Example

```
gui.Image {
    w = 4,
    h = 2,

    -- Texture to draw.
    texture_name = "texture.png",
    middle_x = 3, -- Optional
    middle_y = 4, -- Optional
    middle_x2 = 5, -- Optional
    middle_y2 = 6, -- Optional
}
```

`gui.ImageButton`

A button that is drawn as a picture instead of the usual button face. The picture (`texture_name`) is stretched to `w` by `h` , and `label` is drawn on top of it. It works like `gui.Button` otherwise.

- `noclip` : Allow the button to be drawn outside the form's area.
- `drawborder` : Draw the normal button border around the picture.
- `pressed_texture_name` : Picture to show while the button is held down.

Example

```
gui.ImageButton {
    w = 6,
    h = 2,

    -- Picture to draw as the button.
    texture_name = "texture.png",
    name = "my_image_button", -- Optional
    label = "Hello world!", -- Optional
    noclip = false, -- Optional
    drawborder = false, -- Optional
    pressed_texture_name = "texture.png", -- Optional
}
```

`gui.ImageButtonExit`

Like `gui.ImageButton` , but the form also closes when the button is clicked.

Example

```
gui.ImageButtonExit {
    w = 6,
    h = 2,

    -- Picture to draw as the button.
    texture_name = "texture.png",
    name = "my_image_button_exit", -- Optional
    label = "Hello world!", -- Optional
    noclip = false, -- Optional
    drawborder = false, -- Optional
    pressed_texture_name = "texture.png", -- Optional
}
```

`gui.ItemImage`

Shows the picture that is used for an item or node in inventories (`item_name` , such as `"default:dirt"`), stretched to fit `w` by `h` .

Example

```
gui.ItemImage {
    w = 2,
    h = 3,

    -- Registered item or node name.
    item_name = "Hello world!",
}
```

`gui.ItemImageButton`

A button that shows the inventory picture of an item or node (`item_name`). By default, the item's description is shown as a tooltip when the player hovers over the button, which you can replace with the `tooltip` parameter. It works like `gui.Button` otherwise.

Give every item image button its own unique `name` , otherwise its tooltip may not work.

Example

```
gui.ItemImageButton {
    w = 1,
    h = 2,

    -- Registered item or node name.
    item_name = "Hello world!",
    name = "my_item_image_button", -- Optional
    label = "Hello world!", -- Optional
}
```

`gui.Label`

Shows text. Labels do not wrap long lines automatically: use `\n` to start a new line and do not use labels for long paragraphs. For those, use `gui.Hypertext` , or a `gui.Textarea` without a `name` (which wraps its text and gets a scrollbar if the text is too long).

Labels are sized to fit their text, and flow ignores color escape sequences when it estimates the width. To make the text bigger or smaller, use the `font_size` style (for example `style = {font_size = "*2"}`), see Styling.

Example

```
gui.Label {
    label = "Hello world!",
}
```

`gui.List`

Shows a grid of inventory slots, with the items of one inventory list in it. This is how you show the player's inventory, a chest, a furnace's input slots, and so on.

- `inventory_location`: Which inventory to show. Common values are `"current_player"` (the inventory of the player who has the form open), `"nodemeta:X,Y,Z"` (the inventory of the node at that position, for example `"nodemeta:10,5,-3"`), and `"detached:name"` (a detached inventory). `"player:name"` is the inventory of a specific player and is only ever sent to that player.
- `list_name`: The name of the list inside that inventory, such as `"main"` or `"craft"`.
- `w` and `h`: How many slots wide and high to draw (whole numbers).

Slots are filled from left to right, then top to bottom, and the player can move items in and out of them. The list is only shown if the inventory has been sent to the player's client. Only as many slots as the inventory list has are drawn, even if `w` times `h` is larger.

Each slot is 1 by 1, with 0.25 between slots, so a list that is 8 slots wide and 4 high is 9.75 by 4.75 big. To change the slot size or spacing, use the `size` and `spacing` styles, see Styling.

Example

```
gui.List {
    -- Which inventory to show.
    inventory_location = "Hello world!",

    -- Which list in that inventory to show.
    list_name = "Hello world!",
    w = 1,
    h = 2,

    -- Index of the first slot to draw. Unlike most numbers here, this starts
    -- at 0. Default 0.
    starting_item_index = 3, -- Optional
}
```

`gui.Listcolors`

Sets the colors of inventory slots. It is invisible, takes up no space, and should come before the `gui.List` elements it is for.

- `slot_bg_normal`: Background color of a slot.
- `slot_bg_hover`: Background color of a slot while the mouse is over it.
- `slot_border`: Color of the border of the slots.
- `tooltip_bgcolor`: Default background color of item tooltips.
- `tooltip_fontcolor`: Default text color of item tooltips.

Only the first two are required. To use pictures instead of colors for slots, see Features flow does not have.

Example

```
gui.Listcolors {
    -- Slot background color.
    slot_bg_normal = "#FF0000",

    -- Slot background color when hovered.
    slot_bg_hover = "#FF0000",

    -- Slot border color.
    slot_border = "#FF0000", -- Optional

    -- Default tooltip background color.
    tooltip_bgcolor = "#FF0000", -- Optional

    -- Default tooltip text color.
    tooltip_fontcolor = "#FF0000", -- Optional
}
```

`gui.Listring`

Sets where items go when the player shift-clicks them. Every `gui.Listring{inventory_location = ..., list_name = ...}` adds a list to a ring, and shift-clicking an item in one list of the ring moves it to the next list in the ring. The first time a list appears in the ring decides which inventory the items are sent to.

`gui.Listring{}` with no parameters adds the last two `gui.List` elements that came before it. It is invisible and takes up no space.

Example

```
gui.Listring {
    -- Same as for gui.List.
    inventory_location = "Hello world!", -- Optional

    -- Same as for gui.List.
    list_name = "Hello world!", -- Optional
}
```

`gui.Model`

Shows a 3D model that the player can turn around with their mouse.

- `mesh`: File name of the model, from the `models` folder of any loaded mod.
- `textures`: The textures for the model, one for each of the model's materials in order.
- `rotation_x` and `rotation_y`: How far the model is turned at first, in degrees.
- `continuous`: Keep the model rotating by itself.
- `mouse_control`: Whether the player can rotate the model with their mouse. Defaults to `true`.
- `frame_loop_begin` and `frame_loop_end`: The range of animation frames to play. Defaults to all frames.
- `animation_speed`: Frames per second to play the animation at. Defaults to `0` (no animation).

`name` is optional and only useful for styling.

Example

```
gui.Model {
    w = 5,
    h = 3,

    -- Only useful for styling.
    name = "my_model", -- Optional

    -- Model file name.
    mesh = "Hello world!",

    -- One texture for each material of the model.
    textures = { "texture.png", ... },

    -- Starting rotation in degrees.
    rotation_x = 3, -- Optional

    -- Starting rotation in degrees.
    rotation_y = 4, -- Optional

    -- Keep rotating by itself. Default false.
    continuous = false, -- Optional

    -- Let the player rotate the model. Default true.
    mouse_control = false, -- Optional

    -- First animation frame to play.
    frame_loop_begin = 5, -- Optional

    -- Last animation frame to play.
    frame_loop_end = 6, -- Optional

    -- Frames per second. Default 0.
    animation_speed = 7, -- Optional
}
```

`gui.Pwdfield`

A single-line text box for passwords: what the player types is hidden. It works like `gui.Field`, apart from not having a `default`. `ctx.form[name]` is the text that was typed. `on_key_enter` is supported, see Handling enter keypresses in fields.

Example

```
gui.Pwdfield {
    w = 6, -- Optional
    h = 2, -- Optional
    name = "my_pwdfield", -- Optional
    label = "Hello world!", -- Optional

    -- Called when the player presses Enter.
    on_key_enter = function(player, ctx) end, -- Optional
}
```

`gui.ScrollContainer`

An area of a fixed size that clips the elements inside it and moves them when a scrollbar is moved. You usually want `gui.ScrollableVBox` instead, which does all of this for you. The elements inside are positioned by hand with `x` and `y`, like in `gui.Container`.

- `w` and `h`: The size of the visible area.
- `scrollbar_name`: The `name` of the `gui.Scrollbar` that controls the area. You need to add that scrollbar yourself.
- `orientation`: `"vertical"` (the default) or `"horizontal"`.
- `scroll_factor`: How far the contents move for each step of the scrollbar's value. Default `0.1`. It cannot be negative.

Flow cannot work out the range of the scrollbar for you here. Set `max` and `thumbsize` with `gui.ScrollbarOptions` before the scrollbar, where `max` is the height that is hidden divided by `scroll_factor`.

Example

```
gui.HBox {
    gui.ScrollContainer {
        w = 4, h = 3,
        scrollbar_name = "my_scrollbar",
        orientation = "vertical",
        { type = "label", x = 0, y = 0.5, label = "Top" },
        { type = "label", x = 0, y = 7.5, label = "Bottom" },
    },

    -- The contents are 8 high and 3 are visible, so 5 are hidden. This
    -- is 50 scroll steps of 0.1 each.
    gui.ScrollbarOptions { opts = { max = "50", thumbsize = "19" } },
    gui.Scrollbar {
        w = 0.5, h = 3,
        orientation = "vertical",
        name = "my_scrollbar",
    },
}
```

`gui.Scrollbar`

A slider that the player can drag. `ctx.form[name]` is its position, a whole number from `min` to `max` (by default `0` to `1000`, change them with `gui.ScrollbarOptions` before the scrollbar). `value` is the position it starts at, and `orientation` is `"vertical"` or `"horizontal"`. Use it with a `gui.ScrollContainer`, or by itself as a slider.

`scrollbar_bg`, `slider`, `arrow_up`, and `arrow_down` are textures that replace the parts of the scrollbar. They are not described in the Luanti documentation and may be ignored.

Example

```
gui.Scrollbar {
    w = 8,
    h = 2,

    -- "vertical" or "horizontal".
    orientation = "vertical",
    name = "my_scrollbar", -- Optional

    -- Starting position.
    value = 3, -- Optional

    -- Texture. Optional, see above.
    scrollbar_bg = "texture.png", -- Optional

    -- Texture. Optional, see above.
    slider = "texture.png", -- Optional

    -- Texture. Optional, see above.
    arrow_up = "texture.png", -- Optional

    -- Texture. Optional, see above.
    arrow_down = "texture.png", -- Optional
}
```

`gui.ScrollbarOptions`

Sets the options of the `gui.Scrollbar` elements that come after it. It is invisible and takes up no space. `opts` is a table with any of these options, all values are strings that contain whole numbers (except `arrows`):

- `min`: The smallest value (default `0`).
- `max`: The largest value (default `1000`). If it is the same as `min`, the scrollbar is disabled.
- `smallstep`: How much the value changes when an arrow is clicked or the mouse wheel is scrolled (default `10`). A negative number resets it to `10`.
- `largestep`: How much the value changes for Page Up and Page Down (default `100`). A negative number resets it to `100`.
- `thumbsize`: The size of the thumb (the part that is dragged), counted in values out of the range of the scrollbar. For a scrollbar from 1 to 100, a `thumbsize` of `10` makes the thumb a tenth of the scrollbar long. If it is zero or less, it is reset to `1`.
- `arrows`: `"show"` or `"hide"` the arrow buttons. `"default"` hides them if the scrollbar gets too small, and shows them otherwise.

Example

```
gui.ScrollbarOptions {
    opts = { min = "0", max = "100", smallstep = "5", arrows = "hide" },
}
```

`gui.SetFocus`

Gives keyboard focus to the element with this `name`. Put it before that element. Normally the focus is not changed when the form is shown again (for example by `form:update()`), so that the focus the player chose is kept. Set `force` to `true` to move the focus every time.

Checkboxes, buttons (of all kinds), tables, text lists, dropdowns, fields, password fields, text areas, and scrollbars can be focused.

Example

```
gui.SetFocus {
    -- Name of the element to focus.
    name = "my_set_focus",

    -- Move the focus every time. Default false.
    force = false, -- Optional
}
```

`gui.Style`

Sets style properties of elements by their `name`. See Styling. It is invisible and takes up no space.

Example

```
gui.Style {
    selectors = { "my_button" },
    props = { bgcolor = "red", textcolor = "yellow" },
}
```

`gui.StyleType`

Sets style properties of every element of a type. See Styling. It is invisible and takes up no space.

Example

```
gui.StyleType {
    selectors = { "button", "image_button" },
    props = { textcolor = "yellow" },
}
```

`gui.Tabheader`

A row of tabs. `captions` is the text of each tab, `ctx.form[name]` is the number of the selected tab (the first tab is `1`), and `current_tab` is the tab that is selected at first. Showing the right contents for the selected tab is up to you, usually by reading `ctx.form[name]` in your build function.

Give it both a `w` and an `h`: flow cannot work out how much space it needs. If `transparent` is `true` the tabs are semi-transparent, and `draw_border` draws a thin line along the bottom of the tabs.

Example

```
gui.Tabheader {
    h = 1,
    name = "my_tabheader", -- Optional

    -- Text of each tab.
    captions = { "Hello world!", ... },

    -- Tab selected at first, the first is 1.
    current_tab = 2, -- Optional

    -- Draw semi-transparent. Default false.
    transparent = false, -- Optional

    -- Draw a line along the tabs. Default false.
    draw_border = false, -- Optional
    w = 3,
}
```

`gui.Table`

A table with rows and columns that scrolls if it has too many rows. The player can select one row, and `ctx.form[name]` is the number of the selected row (the first row is `1`). `selected_idx` is the row that is selected at first.

`cells` is one flat list with the text of every cell, going left to right along each row and then down to the next row. The number of columns comes from `gui.TableColumns` (the default is one text column). Put `gui.TableColumns` and `gui.TableOptions` before the table, as they only affect the table that comes after them. They do not take up any space.

Tables have no natural size, so you must give them a `w` and `h`.

Example

```
gui.Table {
    w = 3,
    h = 2,
    name = "my_table", -- Optional

    -- Text of every cell, row by row.
    cells = { "Hello world!", ... },

    -- Row selected at first, the first row is 1.
    selected_idx = 3, -- Optional
}
```

Example with columns and options

```
gui.VBox {
    gui.TableColumns {
        tablecolumns = {
            { type = "text", opts = { width = "8" } },
            { type = "text", opts = { align = "right" } },
        },
    },
    gui.TableOptions {
        opts = { highlight = "#335577" },
    },
    gui.Table {
        w = 10, h = 4,
        name = "scores",
        cells = {
            "Alice", "10",
            "Bob", "7",
        },
    },
}
```

`gui.TableColumns`

Sets the columns of the `gui.Table` that follows it. `tablecolumns` is a list with one `{type = ..., opts = {...}}` entry for each column, from left to right. `type` is one of:

- `"text"`: Show the cell contents as text.
- `"image"`: The cell contents are a number that picks an image, which you set with the options `["0"] = "texture.png"`, `["1"] = "other.png"`, and so on (the numbers do not have to be consecutive). Empty cells and cells that are not numbers use image `0`. Images are scaled down to fit the height of the row if necessary.
- `"color"`: The cell contents are a color that is used for the text of the next column(s). Use the `span` option to set how many columns it affects (default: all remaining columns).
- `"indent"`: The cell contents are a number that sets how far the next column is indented.
- `"tree"`: Like `"indent"`, but the player can open and close the indented rows, like in a file browser.

`opts` is a table of options, all values are strings:

- `align`: For text and image columns, where the content goes in the cell: `"left"` (default), `"center"`, `"right"`, or `"inline"`.
- `width`: For text and image columns, the minimum width, in "em" units (relative to the font size, default `0`). For indent and tree columns, how far each level is indented (default `1.5`).
- `padding`: Space to the left of the column, in "em" units (default `0.5`, or `0` for indent columns).
- `tooltip`: Text to show when hovering over the column.

Example

```
gui.TableColumns {
    tablecolumns = {
        { type = "text", opts = { width = "8" } },
        { type = "text", opts = { align = "right" } },
    },
}
```

`gui.TableOptions`

Sets the appearance of the `gui.Table` that follows it. `opts` is a table with any of these options, all values are strings:

- `color` : Default text color (default `"#FFFFFF"`).
- `background` : Background color of the table (default `"#000000"`).
- `border` : `"true"` or `"false"` , whether to draw a border (default `"true"`).
- `highlight` : Background color of the selected row (default `"#466432"`).
- `highlight_text` : Text color of the selected row (default `"#FFFFFF"`).
- `opendepth` : How many levels of `"tree"` column rows are open at first. Rows deeper than this are collapsed (default `"0"`).

Example

```
gui.TableOptions {
    opts = {
        background = "#222222",
        highlight = "#335577",
        border = "false",
    },
}
```

`gui.Textarea`

A multi-line text box. Long lines are wrapped to fit the width, and a scrollbar is added if the text is too tall. `label` , if set, is shown above the box and `default` is the text it starts with. `ctx.form[name]` is the text in the box, including its newlines.

If you leave out `name` , the text cannot be edited and no box is drawn, which makes it a label that wraps text and scrolls. Because it wraps, you should set a `w` and `h` for this.

Example

```
gui.Textarea {
    w = 6, -- Optional
    h = 2, -- Optional
    name = "my_textarea", -- Optional
    label = "Hello world!", -- Optional

    -- Text the box starts with.
    default = "Hello world!", -- Optional
}
```

`gui.Textlist`

A list of text rows that scrolls if there are too many to fit. The player can select a row, and `ctx.form[name]` is the number of the selected row (the first row is `1`). `selected_idx` is the row that is selected at first.

`listelems` is the text of each row. To color a row, start it with `#RRGGBB` (six hexadecimal digits). To start a row with a literal `#`, write `##`. If `transparent` is `true`, no background is drawn behind the list.

Example

```
gui.Textlist {
    w = 1, -- Optional
    h = 2, -- Optional
    name = "my_textlist", -- Optional

    -- Text of each row.
    listelems = { "Hello world!", ... },

    -- Row selected at first, the first row is 1.
    selected_idx = 3, -- Optional

    -- Do not draw a background. Default false.
    transparent = false, -- Optional
}
```

`gui.Tooltip`

Sets the text that is shown when the player hovers over another element, and optionally the colors of that tooltip. `gui_element_name` is the `name` of that element. Put the tooltip after the element it belongs to. It is invisible and does not take up any space.

If you only need to add some text, use the `tooltip` parameter on the element itself, see Tooltips.

Example

```
gui.Tooltip {
    -- Text to show.
    tooltip_text = "Hello world!",

    -- Background color of the tooltip.
    bgcolor = "#FF0000", -- Optional

    -- Text color of the tooltip.
    fontcolor = "#FF0000", -- Optional

    -- Name of the element to attach to.
    gui_element_name = "my_button",
}
```

`gui.Vertlabel`

Shows text vertically, with one letter below the other, from top to bottom.

Example

```
gui.Vertlabel {
    label = "Hello world!",
}
```

3.3 Hypertext markup

`gui.Hypertext` shows text that is written in a small markup language with tags that look like HTML tags. It can be used to make text bold, coloured, bigger, or centred, to add pictures, and to add clickable links.

Warning: The markup language is currently unstable and may change. Use it with caution. Flow passes the text to Luanti without looking at it, so everything on this page is handled by the player's Luanti client, not by flow.

```
gui.Hypertext {
    w = 8, h = 3,
    text = "<center><big>Welcome!</big></center>" ..
        "This is <b>bold</b>, <i>italic</i>, and " ..
        "<style color='red'>red</style> text.",
}
```

What flow does not have: the markup can only be used in `gui.Hypertext`. Luanti can also show it in a tooltip (`hypertip[]`), but flow has no element for that, see Features flow does not have. Colors in the markup are written like colors everywhere else, see Colors.

3.3.1 Syntax

Some tags enclose text: they open with `<tagname>` and close with `</tagname>`. Tags can have attributes, which are written in the opening tag as `key=value`. Attribute values should be quoted with either `"` or `'`. Inside a Lua string that is already in double quotes, use single quotes, as in `<style color='red'>`.

To write a literal `<`, `>`, or `\` in the text, put a backslash before it. In a quoted attribute value, put a backslash before a literal quote mark. In Lua code, each backslash has to be written twice (`"\\<"`), or you can use a long string (`[[\<]]`).

3.3.2 Basic tags

`<style color=... font=... size=...>...</style>`

Changes the style of the text inside it.

- `color`: Text color.
- `size`: Text size.
- `font`: Text font, `mono` or `normal`.

`<global ...>`

Sets the style of the whole text. You only need to use it once. If you put several in the text, their changes are made in the order the tags appear.

Attributes that only apply to the whole text:

- `background` : Background of the text, a color or `none` .
- `margin` : Space around the edge of the text, in pixels.
- `valign` : Vertical alignment of the text: `top` , `middle` , or `bottom` .

Attributes that are inherited by everything inside the text:

- `color` : Default text color.
- `hovercolor` : Color of `<action>` links while the mouse is over them.
- `size` : Default text size.
- `font` : Default font, `mono` or `normal` .
- `halign` : Default horizontal alignment: `left` , `right` , `center` , or `justify` .

`<tag name=... color=... hovercolor=... font=... size=...>`

Defines a new tag, or changes the style of an existing one.

- `name` : The name of the tag to define or change.
- `color` : Text color.
- `hovercolor` : Text color while the mouse is over it (only for `action` tags).
- `size` : Text size.
- `font` : Text font, `mono` or `normal` .

3.3.3 Layout tags

These tags are always available. You can add more with `<tag ...>`.

Tag	Effect
`<normal>...</normal>`	Normal size text
`<big>...</big>`	Big text
`<bigger>...</bigger>`	Bigger text
`<center>...</center>`	Centred text
`<left>...</left>`	Left-aligned text
`<right>...</right>`	Right-aligned text
`<justify>...</justify>`	Justified text
`<mono>...</mono>`	Monospaced font
`<b>...</b>`, `<i>...</i>`, `<u>...</u>`	Bold, italic, and underlined text

3.3.4 Links

`<action name=...>...</action>`

Makes the text inside it clickable. `name` is required. An optional `url` attribute is a web address to open when the text is clicked.

When the player clicks it, the form is sent to the server and the value of the `gui.Hypertext` element is `"action:"` followed by the action name. In flow, give the element a `name` and an `on_event` function, and read the value from `ctx.form`. This is experimental and may change.

```
gui.Hypertext {
    w = 6, h = 1,
    name = "help",
    text = "Read the <action name='rules'>rules</action>.",
    on_event = function(player, ctx)
        if ctx.form.help == "action:rules" then
            core.chat_send_player(player:get_player_name(), "No rules!")
        end
    end,
}
```

3.3.5 Pictures

`<img name=... float=... width=... height=...>`

Draws a texture that is in the client's media cache.

- `name` : The name of the texture (required).
- `float` : If present, the picture floats to the `left` or `right` and the text flows around it.
- `width` , `height` : Force a size instead of using the size of the texture. If you only give one of them, the shape of the texture is kept.

`<item name=... float=... width=... height=... rotate=... angle=...>`

Draws the inventory picture of an item.

- `name` : The item name (required), such as `default:dirt` .
- `float` , `width` , `height` : Like for `<img>` .
- `rotate` : Make the item turn. Either `yes` or `X,Y,Z` , the rotation speeds around each axis as percentages of the standard speed (-1000 to 1000).
- `angle` : The angle the item is shown at, as `X,Y,Z` , the angles around each of the three axes.

`rotate` and `angle` only work if the player has the `inventory_items_animations` setting turned on.

4. Other features

4.1 Unsupported Features

Flow does not support everything that Luanti formspecs can do. This page lists what is missing, so you know before you look for it. Where there is a way to get a similar result, it is mentioned.

4.1.1 Missing elements

Formspec element	What it does	What to do instead
`hypertip[]`	A tooltip that shows formatted text (see Hypertext markup).	Flow tooltips only show plain text with the `tooltip` parameter or `gui.Tooltip`. The markup language can only be used in `gui.Hypertext`.
`allow_close[]`	Stops the Escape key (and similar) from closing the form.	Nothing. Players can always close a flow form. You can run code when they do with `on_quit`, but you cannot stop it.
`listimages[]`	Pictures for the background of inventory slots.	`gui.Listcolors` only sets colors.
`label[x,y;w,h;text]`	A label that wraps long text inside a given size (an "area label").	`gui.Textarea` without a `name`, or `gui.Hypertext`. Because of this, the `halign` and `valign` styles do not do anything for `gui.Label`.
`tooltip[x,y;w,h;...]`	A tooltip for an area instead of an element.	The `tooltip` parameter of any element (flow uses the area of an element without a `name` for you). `gui.Tooltip` can only be attached to a named element.
`field[name;label;default]`	A small form with a single field and an automatic "Proceed" button (for example for signs).	Use a `gui.Field` and a `gui.Button`.

4.1.2 Missing parameters

- **Content padding of scroll containers.** Luanti can work out the range of a scroll container's scrollbar for you, but flow's `gui.ScrollContainer` does not expose this, so you have to set the range yourself with `gui.ScrollbarOptions`. `gui.ScrollableVBox` already does this for you.

- **Node formspec inventory locations.** The `"current_node"` inventory location refers to the node whose formspec is open. A flow form cannot be the formspec of a node, so use `"nodemeta:X,Y,Z"` to show a node's inventory instead.

4.1.3 Things flow does for you

These formspec elements exist, but flow generates them itself and you cannot write them:

- `size[]`: The size of the form is worked out from the elements in it.
- `position[]`, `anchor[]`, and `padding[]`: Use the `window_position`, `window_anchor`, and `window_padding` options of the root element, see Experimental features.
- `no_prepend[]` and `bgcolor[]`: Use the `no_prepend`, `bgcolor`, `fbgcolor`, and `bg_fullscreen` options of the root element.
- `real_coordinates[]`: Flow always uses the newer coordinate system, where there is no hidden padding or spacing between elements.
- `container_end[]` and `scroll_container_end[]`: Elements inside `gui.Container` and `gui.ScrollContainer` are written inside the table.
- `formspec_version[]`: Flow always uses formspec version 7.

4.1.4 Newer Luanti features

Because flow always uses formspec version 7 (Luanti 5.8), features that were added in newer versions may not work, even if the missing element or parameter is not listed above. This includes the area label (version 9), floating-point animation frames in `gui.Model` (version 10), and the `halign` and `valign` styles of fields and text areas (version 11), and the `editable` style of checkboxes, dropdowns, and fields (version 12, the only addition in that version). Formspec versions are described in the "Formspec Version History" section of the Luanti API documentation.

4.2 Styling forms

Styles change how elements look: their colors, images, fonts, and so on. A style is a set of properties such as `bgcolor = "red"`, and which properties exist depends on the type of element. All of them are listed in the property reference below.

You can style a single element with the `style` field (the inline syntax), or style many elements at once with the `gui.Style` and `gui.StyleType` elements (see separate style elements).

4.2.1 Inline syntax

At the moment I suggest only using this syntax if your form won't look broken without the style - older versions of flow don't support this syntax, and I may make breaking changes to the sub-style syntax in the future.

You can add inline styles to elements with the `style` field:

```
gui.Button {
    label = "Test",
    style = {
        bgcolor = "red",

        -- You can style specific states of elements:
        { sel = "$hovered", bgcolor = "green" },

        -- Or a combination of states:
        { sel = "$hovered, $pressed", bgcolor = "blue" },
        { sel = "$hovered+pressed", bgcolor = "white" },
    },
}
```

The properties directly inside `style` always apply. A `sel` entry only applies while the element is in one of the listed states. Separate states with a comma to apply the style when *any* of them is active (`"$hovered, $pressed"`), or join them with `+` to apply it only when *all* of them are active at once (`"$hovered+pressed"`).

If you need to style multiple elements, you can reuse the `style` table:

```
local my_style = { bgcolor = "red", { sel = "$hovered", bgcolor = "green" } }

local gui = flow.make_gui(function(player, ctx)
    return gui.VBox {
        gui.Button { label = "Styled button", style = my_style },
        gui.Button { label = "Unstyled button" },
        gui.Button { label = "Second styled button", style = my_style },
    }
end)
```

Note that this may inadvertently reset styles on subsequent elements if used on elements without a name. Luanti can only style unnamed elements by their type, so flow applies the style to every element of that type and then resets those properties to their defaults straight after the element. This also undoes any earlier `gui.StyleType` for the same properties. Give the element a `name` to avoid this.

Property values are normally strings, but flow also accepts numbers and booleans where a property needs one. Setting a property to an empty string (`bgimg = ""`) resets it to its default value.

4.2.2 Separate style elements

Alternatively, you can use the `gui.Style` and `gui.StyleType` elements if you need to style a large group of elements or need to support older versions of flow:

```
gui.Style {
    selectors = { "btn1" },
    props = {
        bgimg = "button.png",
        border = false,
    }
},

gui.Button {
    name = "btn1",
    label = "Button",
},
```

The `Style` and `StyleType` elements are invisible and won't affect padding. They only affect the elements that come after them, so put them before the elements you want to style.

Warning: `gui.Style` and `gui.StyleType` do not support styles that change the element size, such as `font_size` for labels, buttons, and fields, and `size` / `spacing` for lists. Flow only sees these properties in the inline syntax, so use that for them.

4.2.3 Selectors and states

`selectors` is a list that says which elements the style applies to:

- `gui.Style` selects elements by their `name` , so `{"btn1"}` styles the element with `name = "btn1"` .
- `gui.StyleType` selects elements by their type, so `{"button"}` styles every button that comes after it. The type is the element's name in lowercase with underscores between words, for example `item_image_button` for `gui.ItemImageButton` . Some types also style other types, see element types.
- `gui.StyleType` also accepts `"*"` , which selects every element.

A selector can end with a colon and one or more states to only style the element while it is in that state, such as `"btn1:hovered"` or `"button:hovered+pressed"` (`+` means all of the states have to be active).

```
gui.StyleType {
    selectors = { "button", "image_button" },
    props = { textcolor = "yellow" },
},
gui.StyleType {
    selectors = { "button:hovered" },
    props = { bgcolor = "#336699" },
},
```

States

Every element has the `default` state, which is the same as not giving any state. Buttons (`gui.Button` , `gui.ButtonExit` , `gui.ImageButton` , `gui.ItemImageButton`) also have these states:

State	Active when
`focused`	The button has keyboard focus
`hovered`	The mouse is over the button
`pressed`	The button is being pressed

Element types

These are the types that can be styled. A type that "inherits" from another also uses the styles set for the other type, for example a style set on `image` also applies to `animated_image`.

Type	Element	Inherits from
`animated_image`	`gui.AnimatedImage`	`image`
`box`	`gui.Box`	
`button`	`gui.Button`	
`button_exit`	`gui.ButtonExit`	`button`
`checkbox`	`gui.Checkbox`	
`dropdown`	`gui.Dropdown`	
`field`	`gui.Field`	
`image`	`gui.Image`	
`image_button`	`gui.ImageButton`	
`item_image_button`	`gui.ItemImageButton`	
`label`	`gui.Label`	
`list`	`gui.List`	
`model`	`gui.Model`	
`pwdfield`	`gui.Pwdfield`	`field`
`scrollbar`	`gui.Scrollbar`	
`tabheader`	`gui.Tabheader`	
`table`	`gui.Table`	
`textarea`	`gui.Textarea`	
`textlist`	`gui.Textlist`	
`vertlabel`	`gui.Vertlabel`	`label`

Luanti also lists `hypertip`, but flow has no element for it, see Features flow does not have. `gui.ItemImage` has the `item_image` type, which only supports `noclip` (see Other elements). `gui.ImageButtonExit` and `gui.ButtonURL` are not in Luanti's list of types, so give them a `name` and style them with `gui.Style` or the inline syntax.

4.2.4 Property reference

Colors are described in Elements, and textures are image file names. A "2D vector" is written as a string such as `"1,2"`. Properties that only list a type of element are not valid on other elements, and an unknown property is ignored.

All elements

Property	Description
`noclip`	Boolean. Set to `true` to let the element be drawn outside the area of the form. Default `false`.

Text properties

These properties change how text looks. `font` and `font_size` are valid on buttons, labels, fields, tables, and text lists. `textcolor` is valid on buttons, labels, fields, and tab headers, but not on tables and text lists. The sections below show which applies where.

Property	Description
`font`	Font type, a comma-separated list. Use at most one of `normal` (the default font) and `mono` (a font where every letter is the same width), and any of `bold` and `italic`. For example `"mono,bold"`. If you only give `bold` or `italic`, `normal` is used. Default `normal`.
`font_size`	Text size. `"16"` sets an absolute size, `"+2"` and `"-2"` make the text 2 points larger or smaller than the default, and `"*1.5"` multiplies the default size by 1.5 (like `em` in CSS). The default is the size the player has chosen in their settings.
`textcolor`	Color of the text. Default white.

Flow can make room for bigger text in `gui.Label`s, buttons, and the default text of fields, but only if `font_size` starts with `*` and is set with the inline syntax (`style = {font_size = "*2"}`). Other values still change the text but not the size of the element. The labels above fields and checkboxes ignore `font_size`, and a `gui.Label` with a fixed `h` is positioned as if `font_size` was not set (this is deprecated).

Buttons

For `gui.Button`, `gui.ButtonExit`, `gui.ImageButton`, and `gui.ItemImageButton`. Buttons also support `font`, `font_size`, and `textcolor`.

Property	Description
`alpha`	Boolean. Whether to draw the transparent parts of `bgimg`. Default `true`.
`bgcolor`	Color that tints the button.
`bgimg`	Texture to use as the background of the button. Default none.
`bgimg_middle`	Draws `bgimg` as a nine-slice. Written as a string in one of the formats `"x"`, `"x,y"`, or `"x,y,x2,y2"`, which have the same meaning as `middle_x`, `middle_y`, `middle_x2`, and `middle_y2`. Also adds padding around the content of the button.
`border`	Boolean. Whether to draw the bevelled button border. Set to `false` to hide it. Default `true`.
`content_offset`	2D vector that moves the content (text and image) of the button without resizing the button.
`padding`	Space between the edges of the button and its content, as a rectangle written in the same formats as `bgimg_middle`. This value is relative to `bgimg_middle`.
`sound`	A sound to play when the button is triggered.

`gui.ImageButton` has some more properties for the picture drawn on top of the button:

Property	Description
`fgimg`	The picture to draw. Default none. The `texture_name` and `pressed_texture_name` you give to the element take precedence over this.
`fgimg_middle`	Draws `fgimg` as a nine-slice, like `bgimg_middle`.

Some older properties still work but are deprecated. Use states instead: `bgcolor_hovered`, `bgcolor_pressed`, `bgimg_hovered`, `bgimg_pressed`, `fgimg_hovered`, and `fgimg_pressed`. They work like `bgcolor` and `bgimg` but only apply while the button is hovered or pressed.

Fields and text areas

For `gui.Field`, `gui.Pwdfield`, and `gui.Textarea`. They also support `font`, `font_size`, and `textcolor`.

Property	Description
`border`	Boolean. Set to `false` to hide the background and border of the text box. Default `true`.
`halign`	Horizontal alignment of the text: `left` (default), `center`, or `right`. Needs a recent version of Luanti (5.17 or newer), and flow may not be able to use it, see Features flow does not have.
`valign`	Vertical alignment of the text: `top` (default), `center`, or `bottom`. Needs a recent version of Luanti (5.17 or newer). Only has an effect when all of the text fits inside the element. If the text is too long (and a `gui.Textarea` gets a scrollbar), it is always aligned to the top so that no text is cut off. It also does not work in `gui.Field`, but `halign` does.

Labels

For `gui.Label` and `gui.Vertlabel`: `font`, `font_size`, and `textcolor`.

Luanti also has `halign` and `valign` for labels, but they only apply to the area label, which flow does not have, so they do nothing here (see Features flow does not have).

Lists

For `gui.List`:

Property	Description
`size`	2D vector. The size of each slot. Default `"1,1"`.
`spacing`	2D vector. The space between slots. Default `"0.25,0.25"`.

Flow uses these values to work out how big a list is, but it only understands whole numbers in them (such as `"2,2"` or `"0,0"`).

Boxes

For `gui.Box`. These properties only take effect when the box has no `color` of its own. The `colors`, `bordercolors`, and `borderwidths` properties can have one, two, or four values separated by commas (like CSS borders):

- One value, such as `"#FF0"`: used for all corners or borders.
- Two values, such as `"red,#FFAAFF"`: for the colors, the top-left and bottom-right corners, then the top-right and bottom-left corners. For the widths, the top and bottom borders, then the left and right borders.
- Four values, such as `"blue,#A0F,green,#FFFA"`: starting with the top-left corner (or the top border) and going clockwise.

Property	Description
`colors`	Color of the corners of the box. Default `black`.
`bordercolors`	Color of the borders of the box. Default `black`.
`borderwidths`	Width of the borders in pixels, as a whole number. A negative width puts the border inside the box, a positive width puts it outside the box, and `0` draws no border. Default `0`.

Models

For `gui.Model`:

Property	Description
`bgcolor`	Background color of the model area.

Read-only elements (`editable`)

Luanti 5.18 (formspec version 12) added an `editable` property for `gui.Checkbox`, `gui.Dropdown`, `gui.Field`, `gui.Pwdfield`, and `gui.Textarea`. Setting it to `false` makes the element read-only (default `true`). Flow does not support it yet, because flow always uses formspec version 7, so it may do nothing or be ignored depending on the player's version of Luanti. To make a field read-only for now, handle that in your own code, for example by ignoring changes to it. See also Newer Luanti features.

Other elements

Element	Properties
`gui.Checkbox`	`sound`: A sound to play when the box is ticked or unticked.
`gui.Dropdown`	`sound`: A sound to play when a different item is selected.
`gui.Tabheader`	`sound`: A sound to play when a different tab is selected. `textcolor`: Color of the tab text, default white.
`gui.Table`, `gui.Textlist`	`font` and `font_size`. The colors of a table are set with `gui.TableOptions`.
`gui.Image`, `gui.AnimatedImage`, `gui.ItemImage`, `gui.Scrollbar`	Only `noclip`.

4.3 Padding, spacing, and backgrounds

All elements can have a `padding` value, which will add the specified amount of padding around the element. The "root" element of the form (the one returned by `build_func`) has a default padding of 0.3, everything else has a default padding of 0.

`HBox` and `VBox` have a `spacing` field which specifies how much spacing there is between elements inside the box. If unspecified, `spacing` will default to 0.2.

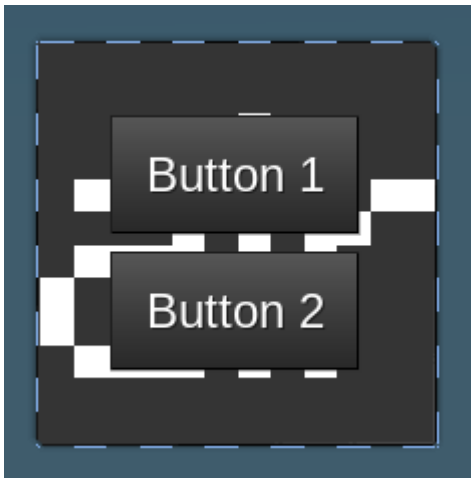
Container elements (HBox and VBox) can optionally have `bgimg` and `bgimg_middle` parameters that specify a background for the container. `bgimg` is the name of a texture. If `bgimg_middle` is set, the texture is drawn as a nine-slice whose corners are `bgimg_middle` pixels big, so one small texture can make a background of any size. The background will be drawn behind any padding that the container has.

Example:

```
gui.VBox {
    padding = 0.5,
    spacing = 0.1,

    -- bgimg can be used without bgimg_middle
    bgimg = "air.png",
    bgimg_middle = 2,

    gui.Button { label = "Button 1" },
    gui.Button { label = "Button 2" },
}
```



The padding around the VBox is 0.5 and the spacing between the buttons inside it is 0.1.

4.4 Tooltips

A tooltip is a small box of text that appears when the player hovers over an element.

You can add tooltips to elements using the `tooltip` field:

```
gui.Image {
    w = 2, h = 2,
    texture_name = "air.png",
    tooltip = "Air",
}
```

This works on elements without a `name` as well. In that case the tooltip covers the area of the element, and the tooltip of a named element that overlaps it takes priority.

There is also a `gui.Tooltip` element which lets you change the background and text color of the tooltip. It refers to the element it belongs to by `name`, and should come after that element. As with `gui.Style`, it is invisible and won't affect padding.

```
gui.VBox {
    gui.Button { name = "my_button", label = "Hover over me" },
    gui.Tooltip {
        gui_element_name = "my_button",
        tooltip_text = "Hello!",
        bgcolor = "#000000",
        fontcolor = "#FFFF00",
    },
}
```

4.5 Hiding elements

Elements inside boxes can have `visible = false` set to hide them from the player. Elements hidden this way will still take up space like with `visibility: hidden;` in CSS.

4.5.1 Example

```
gui.VBox {
    gui.Button { label = "First button" },
    gui.Button { label = "Hidden", visible = false },
    gui.Button { label = "Last button" },
}
```

4.5.2 Alternatives

If you don't want hidden elements to take up any space, see the documentation for `gui.Nil`.

4.6 Using a form as an inventory

> Consider using Sway instead if you want to use flow as an inventory replacement while still having some way for other mods to extend the inventory.

A form can be set as the player inventory, which is the screen the player sees when they open their inventory. Flow internally generates the formspec (the text Luanti uses to describe a menu) and passes it to `player:set_inventory_formspec()`. This will completely replace your inventory and isn't compatible with inventory mods like sfinv (the inventory in Minetest Game, which lets mods add tabs).

```
local example_inventory = flow.make_gui(function(player, context)
    return gui.Label { label = "Inventory goes here!" }
end)
core.register_on_joinplayer(function(player)
    example_inventory:set_as_inventory_for(player)
end)
```

Like with the `show_hud` function, `update*` functions don't do anything, so to update it, call `set_as_inventory_for` again with the new context. If the context is not provided, it will reuse the existing context.

```
example_inventory:set_as_inventory_for(player, new_context)
```

While the form will of course be cleared when the player leaves, if you'd like to unset the inventory manually, call `:unset_as_inventory_for(player)`, analogue to `close_hud`:

```
example_inventory:unset_as_inventory_for(player)
```

This will set the inventory formspec string to `""` and stop flow from processing inventory formspec input.

4.7 Manual positioning

4.7.1 Dynamic element types

This is only recommended for elements inside `gui.Container` (see below) outside of some rare use cases that need it, as it does not support flow's layouting elements.

If you want to generate element types from a variable, you can use `{type = "label", label = "Hello world!"}` instead of `gui.Label{label="Hello world!"}`. The `type` is the name of the `gui` function in lowercase with underscores between words, for example `"item_image_button"` for `gui.ItemImageButton`. HBoxes and VBoxes can be created this way as well (with `type = "hbox"` and `type = "vbox"`), however other layouting elements (such as ScrollableVBox and Spacer) won't work correctly.

An example of this is in `example.lua`.

4.7.2 Manual positioning of elements

You can use `gui.Container` elements to contain manually positioned elements. Every element inside a container has an `x` and `y`, which is the distance of its top-left corner from the top-left corner of the container, in the same units as `w` and `h`. Containers can be nested, in which case the offsets add up. If you do not give a `w` and `h`, flow uses the smallest size that fits all the elements inside.

```
gui.VBox {
    gui.Label { label = "Automatically positioned" },
    gui.Container {
        -- You can specify a width and height if you don't want flow to try and
        -- guess at the size of the container.
        -- w = 3, h = 2,

        -- You may embed most plain element tables (a `type`, an `x` and `y`,
        -- and the same parameters as the matching gui function) in a container
        { type = "box", x = 0, y = 0, w = 1, h = 1, color = "red" },
        { type = "box", x = 0.3, y = 0.3, w = 1, h = 1, color = "green" },
        { type = "box", x = 0.6, y = 0.6, w = 1, h = 1, color = "blue" },

        { type = "label", x = 2, y = 1.1, label = "Manually positioned" }
    },
}
```

Note that you should not nest layouted elements (like `gui.VBox`) inside `gui.Container`.

4.8 Experimental features

These features might be broken in the future.

4.8.1 `no_prepend`

Games and mods can add extra content to the start of every form shown to a player with `player:set_formspec_prepend()`, for example to give all forms the same background or style. This is called a "prepend", and it can be why a form has a background or style you did not expect.

You can set `no_prepend = true` on the "root" element to stop prepends from being applied to a form.

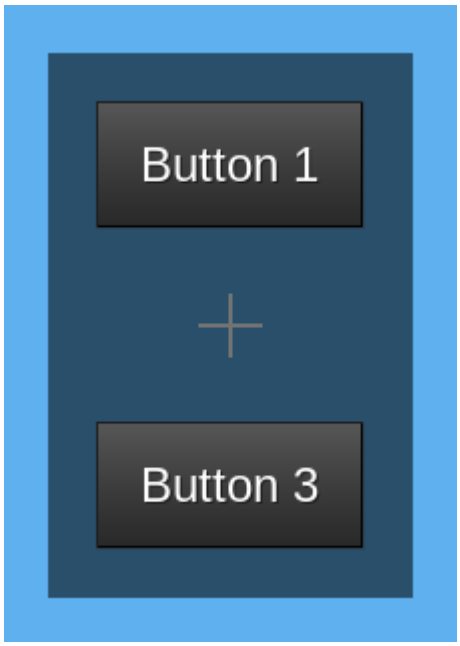
Example:

```
local my_gui = flow.make_gui(function(player, ctx)
    return gui.VBox {
        no_prepend = true,

        gui.Button { label = "Button 1" },

        -- There will be an empty space where the second button would be
        gui.Button { label = "Button 2", visible = false },

        gui.Button { label = "Button 3" },
    }
end)
```



4.8.2 `bgcolor`

You can set `bgcolor = "#123"`, `fbgcolor = "#123"`, and `bg_fullscreen = true` on the root element to set background colors:

- `bgcolor`: The color of the background of the form itself.
- `fbgcolor`: The color of the fullscreen background, which covers the whole screen behind the form.
- `bg_fullscreen`: Which of the two colors to draw. `false` (the default) only draws `bgcolor`, `true` only draws `fbgcolor`, `"both"` draws both, and `"neither"` draws neither.

Colors that you leave out are not changed. See Elements for how colors are written.

4.8.3 Putting the form somewhere else on the screen (likely required for most HUDs)

These values allow the position of the displayed form to be moved around and adjust how it is scaled.

Example:

```
local my_gui = flow.make_gui(function(player, ctx)
    return gui.VBox {
        -- window_position is the point on the screen where the form is
        -- placed, and window_anchor is the point on the form that is put
        -- there. For both, 0 is the top/left and 1 is the bottom/right, and
        -- the default is {x = 0.5, y = 0.5} (the center).
        -- You probably want to set `window_position` and `window_anchor` to
        -- the same value.
        -- This puts the form in the bottom-right corner.
        window_position = { x = 1, y = 1 },
        window_anchor = { x = 1, y = 1 },

        -- Minimum space to leave around the form, as a fraction of the
        -- screen size (here 10% on the left/right and 20% at the top/bottom).
        -- If the form is too big, it will be scaled down. 0 lets the form
        -- touch the edge of the screen and negative values let it go off
        -- the edge.
        -- Default for forms: {x = 0.05, y = 0.05} (i.e. 5% of screen size)
        -- HUDs default to a hardcoded pixel size, if you want them to roughly
        -- line up with formspecs then you may explicitly specify this.
        window_padding = { x = 0.1, y = 0.2 },

        gui.Label { label = "Hello world" },
    }
end)
```

4.8.4 Rendering to a formspec

This API should only be used when necessary and may have breaking changes in the future.

Luanti describes menus with formspec strings, which flow normally generates and shows for you. Some APIs in other mods, such as sfinv, expect formspec strings instead. You can use this API to embed flow forms inside them. To use flow with these mods, you can call `form:render_to_formspec_string(player, ctx, standalone)`.

- A complete formspec starts with a `formspec_version[]` element (the version of the formspec language it uses) and a `size[]` element (the size of the form). By default these aren't included in the returned formspec, and are returned in a third value instead: a table `info` with `w` and `h` (the size of the form, in the units described in Elements) and `formspec_version` (the version number). Set `standalone` to include them in the returned formspec string. The third return value will not be returned.
- Returns `formspec, process_event[, info]`
- The `process_event(fields)` callback will return true if the formspec should be redrawn, where `render_to_formspec_string` should be called and the new `process_event` should be used in the future. This function may return true even if fields.quit is sent.

> **Warning:** Do not use this API with node meta formspecs (formspecs that are stored in the metadata of a node), it can and will break!

4.8.5 Embedding a form into another form

You can embed form objects inside others like this:

```
local parent_form = flow.make_gui(function(player, ctx)
    return gui.VBox {
        gui.Label { label = "Hello world" },
        other_form:embed {
            -- You can optionally pass in the player object to support older
            -- versions of flow (before 2025-06-17).
            player = player,

            -- A name for the embed. If this is specified, the embedded form
            -- will get its own context (accessible at ctx.my_embed_name) and
            -- field names will be rewritten to avoid conflicts with the
            -- parent form. If name is not specified, the embedded form will
            -- share ctx and ctx.form with the parent, and will not have field
            -- names rewritten.
            name = "my_embed_name",
        },
    }
end)
```

Special characters (excluding `-` and `_`) are not allowed in embed names.

4.8.6 Running code when a form is closed

`gui.Container`, `gui.HBox`, `gui.VBox`, and `gui.Stack` elements support an `on_quit` callback which gets run when a player closes a form (for example by pressing Escape or clicking an exit button).

Note that this function is not called in some cases, such as when the player leaves without closing the form or when another form/formspec is shown.

This function must not return anything, behaviour may get added to return values in the future.

```
local parent_form = flow.make_gui(function(player, ctx)
    return gui.VBox {
        on_quit = function(player, ctx)
            core.chat_send_player(player:get_player_name(), "Form closed!")
        end,
    }
end)
```

If multiple `on_quit` callbacks are specified in different elements, they will all get called.

4.8.7 Handling enter keypresses in fields

`gui.Field` and `gui.Pwdfield` support an `on_key_enter` callback that gets called if enter is pressed:

```
local form = flow.make_gui(function(player, ctx)
    return gui.VBox {
        gui.Field {
            label = "Press enter!",
            name = "field",
            on_key_enter = function(player, ctx)
                core.chat_send_player(player:get_player_name(),
                    "Field value: " .. dump(ctx.form.field))
            end,

            -- Pressing enter does not close the form by default. You can
            -- specify close_on_enter to close it when enter is pressed.
            -- This only applies to gui.Field.
            close_on_enter = true,
        },
    }
end)
```

Notes:

- If you're using this callback, please make sure there's some other way to trigger the enter action (like a button) to support older flow versions and in case I replace this API with a better one in the future.
- If you want recent mobile clients to call this callback when editing text, add `enter_after_edit = true` to the field definition. Without it, pressing "Done" on the on-screen keyboard of an Android device does not count as pressing enter. This is experimental.
- The similarly named `on_event` gets called whenever the client submits the field to the server, which could be at any time, and is not very useful, but is still supported for compatibility (and there may be uses for it, such as sanitising field values). Be careful not to accidentally use the wrong callback.

4.8.8 Getting a reference to the current player and context

If you're making a custom widget, it might be useful to get a reference to `ctx` and `player` so you don't have to pass them in manually.

```
local function HelloWorld(def)
    local ctx, player = flow.get_context()
    return gui.Label {
        label = "Hello, " .. player:get_player_name() .. "!\n" ..
            "some_value=" .. ctx.some_value,
        style = def.style,
    }
end

local form = flow.make_gui(function(player, ctx)
    ctx.some_value = 123
    return HelloWorld { style = { font_size = "*2" } }
end)
```

`flow.get_context()` will error when called outside a build function (as you should not normally do this).

4.8.9 `gui.Flow` container

`gui.Flow` is a fixed-width container that wraps items around if they don't fit. Unlike other elements, you must specify a width, and some options like `min_w` and `padding` aren't supported.

This is similar to GTK's `FlowBox`, but with a lot fewer features.

```
gui.Flow {
    -- Fits 3 items
    w = 3 * (1 + 0.2),

    gui.Image { w = 1, h = 1, texture_name = "default_mese_crystal.png" },
    gui.Image { w = 1, h = 1, texture_name = "default_mese_crystal.png" },
    gui.Image { w = 1, h = 1, texture_name = "default_mese_crystal.png" },
    gui.Image { w = 1, h = 1, texture_name = "default_mese_crystal.png" },
    gui.Image { w = 1, h = 1, texture_name = "default_mese_crystal.png" },
}
```

4.9 Popovers (highly experimental)

This API will likely have breaking changes in the future.

Flow supports a highly experimental low-level API for defining popovers. Popovers appear over top of all other elements and are anchored to their parent.

This API is intended to be used for creating themed widgets like dropdowns, or for a higher level popover API that adds styling and is easier to use.

You add a `popover` attribute to any container element to define a popover. Flow does not manage the lifecycle (opening/closing) of popovers, you must ensure that you do this yourself like in the example.

See the below example for details on how to use the API:

```
gui.VBox {
    -- Some padding is added for the purposes of the demo so that the popup
    -- isn't off-screen
    padding = 3,
    gui.Label { label = "Popover demo:" },

    gui.Stack {
        gui.Button {
            label = "Open popover",

            -- When the popover button is clicked, open the popover
            on_event = function(player, ctx)
                ctx.show_popover = true
                return true
            end,
        },

        -- The actual element to overlay
        -- "popover" is set to the gui.VBox element if (and only if)
        -- ctx.show_popover is true, otherwise it's nil so no popover is shown.
        popover = ctx.show_popover and gui.VBox {
            -- Specifying padding and bgcolor is optional, but is probably a
            -- good idea since flow doesn't do any styling on its own.
            padding = 0.2,
            bgcolor = "#222e",

            -- "side" specifies how the popover is positioned relative to the
            -- parent, and can be "bottom" (default), "top", "left", or "right".
            side = ctx.form.side,

            -- align_h and align_v align the popover according to its parent
            -- element. You only need to specify align_h for
            -- side = "top"/"bottom" or align_v for side = "left"/"right",
            -- this example specifies both so that it can demonstrate switching
            -- between different sides.
            align_h = "fill",
            align_v = "center",

            -- Popover contents
            gui.Label { label = "Hi there!" },
            gui.Dropdown {
                name = "side",
                items = {
                    "bottom",
                    "top",
                    "left",
                    "right",
                },
            },
        } or nil,

        -- Clicking outside the popover will call this function, which should
        -- probably close the popover.
        on_close_popover = function(player, ctx)
            ctx.show_popover = false
            return true
        end,
    },

    gui.Label { label = "After popover button" },
}
```

There are some restrictions on popovers:

- They must not extend outside the form, otherwise they'll only be partially visible (unless everything is styled with `noclip = true`, see Styling). Flow does not attempt to detect this.
- You can only define `popover` on container elements, like gui.Stack, gui.HBox, and gui.VBox.
- Only one popover is shown at a time.
- Players can still use tab to interact with things behind the popover, despite being unable to use their mouse to do so.
- You cannot show popovers inside of other popovers.
- Popovers cannot be on the root element.